

ActInf GuestStream #021.1 ~ Adam Safron

GuestStream | Adam Safron | 1:38:12

Hello, everyone. It's April 25th, 2022. We're here with Adam Saffron. And today we're going to be having a guest stream, the radically embodied conscious cybernetic Bayesian brain from free energy to free will and back again. So Adam, thanks a ton for joining on this day. Really looking forward to your presentations. And thanks again, Adam. Take it away.

Thank you for inviting me and letting me come to talk with you. And I'll present some ideas from a recent manuscript in entropy called the radically embodied conscious cybernetic Bayesian brain and free will from free energy to free will and back again. And after that, looking forward to discussing some of the ideas with all of you. And hopefully this is the beginning of an ongoing conversation. So what I tried to do with this manuscript was basically marry good old fashioned cognitive science with a more inactivist perspective, trying to find a middle way between these ways of viewing things. And this is the view that I've come to over a number of years where the idea being that embodiment like at this point, you know, it's not news that the mind is embodied. But the claim is that it is a very, very embodied, some would say radically embodied. I'm not sure if that made the most sense to use that term because the term radical with respect to embodiment is usually pointing to a perspective or radical inactivism that avoids talking about representations or models. And instead of saying, you know, what's really going on is these couplings with the world. And maybe you can say there's like an implicit information processing or intelligence, even that might be objected to. But these talk of representations and symbols and models from cognitive science, that's, those are not real things.

And so this paper goes through a perspective in which basically, we do have representations and models. And we also have these sort of inactive couplings with the world. And I say it's radical and that nearly every mental process you would care to talk about, the way you have to understand it, according to this view, would be first and fundamentally and foremost in terms of its embodied character. So before I move on, is everything coming across clearly? Just okay, sound?

Yeah, I did disable my video because due to some incredible consequences of different situations, I'm using a hotspot, but it's all good.

Okay, you're on the run.

So, let's review a little background. I mean, a lot of this, people who were watching this, but I'll review some background on predictive processing and active inference. Maybe some of what I'll present is somewhat heterodox view. That's also something to discuss. So a basic account of predictive coding, you have these deep pyramidal neurons, and you have these superficial pyramidal neurons, where they're deep with respect to cortex as having this layered structure, roughly six layers with some variations on the theme, depending on where you look. And the idea is that the cortex as a whole is engaging in what's sometimes called hierarchical, or what's often called hierarchical predictive

coding, or potentially predictive processing,

because there might be some assumptions with saying predictive coding that not everyone would sign up for, such as that

you are only suppressing and explaining away information as opposed to sometimes enhancing it.

And there's other nuances that could make people object to saying predictive coding.

Instead of getting into those weeds, the idea would be that you have this hierarchy of beliefs, implicit and neuronal organization.

The nature of these beliefs, and this gets into these debates about representation and inactivism, but this idea that

you could say there's an implicit belief at each neuron and set of neurons about what they expect, what patterns they expect to encounter.

And according to this predictive processing view, through their activity, they're trying to predict this incoming activity and suppress it or explain it away to minimize overall activity.

And that you only update these beliefs where you got it wrong via prediction error.

And this is under certain assumptions. If you have a good model and events in the world are relatively slow relative to your internal signaling units,

you can have substantial energetic savings for an organ of about 2% of your body mass and 20% of your metabolism.

It seems like the kind of thing natural selection would do and there's some decent evidence for this.

So the basics of neural architecture here for Cortex is viewing it as a predictive processing hierarchy or a heterarchy.

And that each of your modalities would be a different belief hierarchy and they would all come together and exchange their information for a joint belief across all your modalities in terms of what you think should be in your sensorium as it moves through the world.

Whether it's the world you're in right now or some sort of counterfactual world in the present or future.

So that's predictive processing in a nutshell.

And so in this account the goal was to use predictive processing to try to get into the micro mechanics of goal oriented behavior and agency.

And so what I suggested is that the work has been done within active inference on this sort of imaginative planning, pursuing of goals.

The work to refer to would be sophisticated inference or sophisticated affective inference, which is thinking about basically planning into the future as this tree search of possibilities where you're going on different branches of these possibilities.

And then picking the ones either in imagination or in reality that would correspond to you minimizing your prediction error the best or realizing value the best.

But this exploration of a tree of possibilities and going down different branches.

And so this would be I suppose a particular flavor of this.

And the idea being that a lot of what would be pulling along this stream of consciousness of imaginings would be this iterative comparison between sensed and imagined states or between various goal states.

So here I have this comic book that a friend helped me make.

And I have multiple comics throughout.

I tend to view comics as being really highly expressive in a way that I think we should probably use them more in terms of being able to capture the joints in experience space with sufficient richness and detail.

And there could even be something like comic book-esque about the stream of experience, which could be part of the reason that they're so effective.

But to return to this comic or a modal frame sequence, as I sometimes call it, you see in the bottom here, this is the actual pose of the agent.

And up top, what's going through the mind's eye, whether we're talking about an imagination or a perception.

And so here it's just a simple contrasting of what you're perceiving in each given moment and then what you're imagining.

So you're sitting, you want tea, you then look and you say, I am not drinking tea.

I am at my computer and there is a laptop in front of me.

You look down and you're imagining having tea in your cup, but then you look and there is indeed no tea.

And then you imagine a next step.

And so what's happening here is you're doing this contrast.

And what I propose is that in contrasting between these states, they're arranging, they're basically setting up prediction errors for each other, where what doesn't match would potentially get you a natural prioritization of what features are unaligned.

And then from those, you use that to retrieve some sort of relevant affordance to help you bridge this gap between your imagination and reality.

And there's a lot of devils in details of the ways this could work, but it's some sort of iterative contrasting of goal states in a goal hierarchy, where it's like you might have an ultimate goal, but to achieve that goal, you have to now go and think of what are all of the sub steps and the sub goals that would be needed to actually get the tea in my hand.

You know, you're going to have to ambulate to the kitchen, make the tea, et cetera, et cetera.

And as you're going back and forth at some place, some point, you either have a plan that's compelling enough or the next action is close enough to your present pose, but you just find yourself launching into action.

Sometimes accompanied by a feeling of having decided, sometimes you just find yourself moving.

But that's the basic account here that the stream of consciousness is being pulled along in this goal oriented fashion, this value driven fashion, sometimes more freely wandering, but sometimes, but, but usually there's an interest.

And you're as an active inference and predictive processing, your preferences or your attracting states, the attractors that you'll find them as a system, which you're attracted to takes the form of predictions.

And so the things in your mind that you're imagining that they become the goal state, they become the state of the world that you prefer in that instance, or in general, if it's a regular kind of imagining.

And this now becomes the attracting state that your system self-organizes and you engage in action selection of different kinds to minimize that difference between what you're imagining and what you are estimating for the world.

So I go into some discussion of neural architecture and different aspects that might be more or less relevant for different aspects of this kind of action selection.

I probably should have just thrown this on top of a picture of a brain, but I didn't, but in some ways it might be a good thing.

And that I wonder whether this account of having a dual tier system, something like a system one and a system two is a broader story than just brains.

And so, you know, I have written out here, like here would be neural substrates of what would enable you to do this.

But, well, it could be across brains.

You might have similar processes happening within, let's say, like an insect nervous system by different means.

Like instead of the basal ganglia, you might be using the mushroom bodies.

If we're talking about like an ant colony, for instance, it's possible you have a different like attracting states of the colony where some are taking this role of like a higher level controller and a lower level controller.

But the basic idea here is that you have an upper level controller that's capable of aggregating information across different.

So you have this lower level process by in this control hierarchy where the lower level is taking care of your immediate couplings with the world.

This would be the more inactive portion.

If we're talking about the brain, it would be something like loosely assembled constellation of cortical reactive dispositions, maybe orchestrated by the cerebellar system of what would be called amortized inference.

And it would be largely unconscious and automatic.

Phenomenologists would talk about just like things being ready at hand as opposed to being present to hand.

You're just effortless engagement and effortless mastery with the world.

But on top of this, you have a slower process.

In this case, I'm saying it's more involved in conscious processing, something more system two like to use Kahneman's language, where it's involved in more deliberate, potentially effortful modes of cognition, which would be a higher level of control over this faster, smooth coupling system.

So go into some details of neural architecture.

Potentially, this might be just what you need for a lot of active inferential agents, something like this across whether or not we're talking about neurons.

This might be a broader story for active inferential agents.

So to return to the example of making tea, go into it some, we're saying that, you know, we're back to some of the neurons.

Here in the brain, areas associated with mental imagery, maybe even physical and computational substrates of mental imagery phenomenality itself,

but like the procunius, for instance, involved in imagination have heavily implicated multiple reasons to think why this would be the case.

But, you know, I have this little Cartesian theater here with the different imaginings.

And down below, the idea is you are iterating through different imaginings, largely via large-scale orchestration by the hippocampal complex.

And it's a capacities to couple with the rest of the cortex to guide these high-level attracting states, where

some portions of which would be your experience.

And so the reason you would bring up the hippocampus in here is it seems to be a specialized subsystem for sequential predictive modeling.

You could say that for cortex as a whole.

Technically, hippocampus is a kind of cortex, archaic cortex, old vintage.

But that basically, so Damasio will talk about this in terms of a convergence-divergence zone, where many things will funnel in and then funnel back with high centrality.

I forget his name.

He's at Waterloo.

The spawn architecture.

But he'll talk about semantic pointers.

But there's different ways of viewing the high centrality of the hippocampus, as it basically will predict this trajectory of the overall organism through space.

You can both encode memories, what was happening in my history, whether in reality or imagination, at each point as I was moving through space and time.

And you could also use it for replay, including in counterfactual circumstances.

And so I have some diagrams here if the neural system is involved.

And right here, you would have the - I think that's good for now.

So if anyone wants to know more details about that, later we could.

So to return to trying to go more phenomenological, so these modal frame sequences are comics.

Ultimately, what I'm trying to build up to is a sufficiently detailed account of neural architecture and a sufficiently detailed account of phenomenology that we can talk about - so that we can basically talk about something as highfalutin as free will and have some purchase. So in this one more complex modal frame sequence, you have that up top, an imagining row. Below that, you have what's actually in your perception.

Below that, you have your pose and your proprioceptive inference.

And then below that, you have your interoceptive inference.

And so as you're stepping through, the stream of consciousness is evolving through space and time, encoded or orchestrated by the hippocampal system at this high - potentially most integrative high level of action selection.

So in this situation, you are having different feelings in your body corresponding to the interoceptive concomitants of - or the interoceptive aspects of what it means to be in different situations.

So for instance, like a hunger situation, here you're getting prediction error, interoceptive prediction error that would be roughly vaguely localized.

Here's like a cool sensation around your gut and mouth or thirst or hunger or something like this.

And now as you're going into movement, you might have here like -

you're still having this interoceptive code of hot and cold states, pleasant and unpleasant, but it's kind of getting infused into other modalities where your proprioception - some of the phenomenology of like feeling like an urge in your body, it's one of the claims I make is that you have this kind of quasi-synesthetic cross-mapping between interoceptive signals and the proprioceptive and exteroceptive - their associates. Part of the reason I claim this and say that this is both part of the phenomenology of will and maybe part of the mechanism also - is that interoceptive signals are poorly localizable in space and time. They tend to have pretty low resolution for them.

And so they may drift free from their sources and as a kind of anomalous inference, but an adaptively anomalous inference, they may get infused into the relevant effector and sensor systems and objects in the worlds for which they would be relevant.

You know, I make a lot of different claims in this work from a number of years of study. A lot of it can be taken a la carte.

Like you can find some things to be more compelling, some things to be less compelling.

But I do think you get a whole that's greater than some of its parts, but it can also be taken in a piecemeal fashion.

So you might not be particularly compelled by this account of interoception as being synesthetic in nature, but nothing really - not too much will ride on that, but it - not too much of what will follow will ride on that, but it could end up being important, so I'm putting it in there.

Okay?

And so across all of this, this is happening within a perspective of computational neurophenomenology. As Neil Seth calls it, or I call it Marian neurophenomenology.

And the reason I call it Marian is specifically his account of multiple levels of analysis, which I guess supervene on each other.

And you can do two things because things have these multiple levels that they can be engaged with.

You can either engage with them a la carte because of - let's say, for instance, you want to do cognitive science,

but you have no idea how the brain works.

And for many years we haven't, and there's a lot of mystery still.

You can kind of bracket.

Like, here's this big set of unknowns, but it's at a different level.

But working at this other level, we can obtain knowledge, even though we might be ignorant at these other levels.

And that would be, you know, functionalism and cognitive science.

But for me, what's more exciting about this, actually, this Marian stack, is the capacity not for independence, but interdependence and mutual constraints.

Using the implementational level or the mechanistic level as a constraint of what's a likely set of algorithms. Using a likely set of algorithms to test hypotheses on the implementation level.

Using what you think the functions of the agents in things like neuropsychiatric literature to constrain - and the neuroimaging literature - to constrain algorithm and thereby implementational.

So the idea of what excites me is, like, filling in this crossword puzzle, leveraging the constraints, and then cross-referencing them with the enriching of phenomena.

So this learning technology is a general approach.

At the other level, I'm using, basically, machine learning as a version of principle.

And typically, going to the races of this idea that a lot of cortex could be understood in terms of particle predictive problem.

But also, evoking some additional machine learning principles.

But that's the overall scope of it.

In terms of the, like, all the body parts and faces, this is part of, like, the radical embodiment idea.

Is that instead of, like, the cortex being this, like, grab bag of associations with areas,

and what does that mean and why,

the idea is that once you really place it front and center as a cybernetic controller for an embodied embedded agent,

once you place embodiment front and center,

a lot of them come up in the experience.

So, for instance, they, like,

okay, there's something like,

like, right around here.

So something like,

but then you're like, wait a second.

That's not too many steps removed from where the ears first pipe in.

So the idea that something like,

you know,

going from phonemes,

or, you know,

going from sound patterns undifferentiated to phonemes and

different steps,

it ends up making cortex and its functions,

in my opinion,

a lot more applicable.

It becomes somewhat like location, location, location,

and location relative to the modalities.

It ends up explaining a lot of things.

It's a similar story for Broca's area.

What a coincidence that it's not too many,

you know,

it's a hop,
skip,
and a jump away from what might move your mouth bits.
There's devils in details here.
I don't want to be glib and underestimated complexity,
but the claim is that actually once you place embodiment in the center,
an elegance kind of falls out and becomes a lot easier to keep track of things.
If you're thinking of cortex as this control,
cybernetic control hierarchy over to basically help a body move through the world
and model the world as it moves through it.
So we are ready,
I think,
to get to free will.
And so taking these modal frame sequences,
and here's doing one for this experience of standing to make tea,
and then taking that model and then pointing it to a Lebet experiment.
So what is a Lebet paradigm?
So some people say Lebet,
some people say Lebet.
I think Lebet said Lebet,
but I'll continue to say Lebet is more common,
and I just will.
But the idea is that you're there,
and you want to see do people,
or don't they have something like meaningful,
conscious control over their actions?
And so the idea is,
okay,
let's strip this down and take like the most insignificant of all actions.
It's like,
well,
surely we need to have control there.
If we're going to say we have control anywhere.
And so the decision to just move your hand based on feeling like you want to do so.
And this coming from basically it's endogenous.
You're not being,
it's not being decided from without,
but you doing something based on something internal to you.

And what was found with the Lebet studies,
or Lebet,
is that you get this ramping activity
preceding the decision to move your hand.
So one more detail.
You have this clock, right?
And there is,
it's a big clock that you're looking at,
and it might be a dot or a big hand moving in a circle.
And the reason that you have this is basically so that you can
estimate the time when it was that you said,
I will move my hand.
When did you decide?
And you then say,
okay,
when you made that decision,
what were you looking at?
Where was this dot as it moved around in a circle?
And this can get you very fine resolution,
sub second resolution,
because if it's a big clock,
that's because then the dot will be moving slowly.
And you can say,
okay,
it was right there.
But what's observed is that you're getting this ramping activity,
which is predictive of the action called the readiness potential.
And it will peak,
you know,
a good 200 milliseconds before you reporting.
It varies,
but it'll peak substantially before you saying,
I've decided.
And so if,
and so the claim is,
well,
if this potential has predictive information of your action,
and is the cause of your action,

and it's happening prior to your conscious decision,
your decision cannot be the thing that caused the action.

There's a,

I mean,

there's a problem right away with this in that with ERP methods,
you're actually averaging over many,
many trials to get these waveforms because you're measuring from the outside of
the brain.

It's like trying to detect like events inside a stadium while measuring,
like listening to the roar of the crowd on the outside.

And you're,

you're limiting what you can do.

So you have to average together a lot of different episodes,
and then you get these event related potentials or ERPs.

And,

but so you get predictive information from this of when people moved,
but the idea that a readiness potential is causal or deterministic in some
fashion,

there's some debate about that,

whether you can infer that I'm going to say that the readiness potential is in
fact,

reflecting a causal process.

And it's reflecting a conscious causal process,

even that is contributing to the action upstream of the decision itself,
where the decision could have some more,

you know,

depends on,

I don't know if there's one account,

but it could have some epiphenomenal aspects to it.

And that the decision is not necessarily the thing.

it's,

it's,

it's not the only a difference maker there,

but the idea is that this is a function of mental imagery,

where you are imagining,

will,

won't move.

And as you're rehearsing,

moving your hand,
you're having different things associated with it.
And that building up of that recurrent,
a building up of activity in these body maps,
whether we're talking about the getting ready to actually change your
proprioceptive pose,
as you're imagining,
starting to move closer to a threshold where you're going to get robust enough
activity that is actually going to drive your muscles and your effector
systems,
or a building of activity in your interoceptive modalities,
like your insular and cingulate hierarchies,
and maybe they're cross,
maybe they're synesthetically cross couplings with other modalities,
that that building up of a feeling of urge is itself consciousness leading to
the stream,
the causal stream of action.

So that I would say,
so yeah,
the,
and then on top of all the different imaginings would be orchestrated by the
capital system would be the basic model.

So
currently,
and so,
and,
and so currently I am working on expanding this to seeing what different
states of consciousness,
the ways in which they could impact these sorts of phenomena,
in terms of changing your awareness of this rehearsal process,
or changing,
that would be one of the primary things of interest.

So,
and also whether differences in the mechanisms of agency could potentially contribute to the
phenomenology of psychedelics.

I'll try to explain what I mean.

So here would be someone relatively unaltered,
engaging in mental rehearsal,

their minds sort of wandering off to nature,
they're seeing a tree,
here's imagination,
here's perception here,
more vivid than imagination.

And they're just sort of imagining things,
imagining,
will I move,
won't I move?

And then at a certain point,
imagination becomes vivid enough that they can really see it happening.
And at a certain point of vividness of imaginings as a function of this rehearsal,
of will I move,
won't I move?

You get enough buildup of whether we're on the implementation level,
we might talk,
or mechanistic level,
we might talk about recurrent activity.

In terms of active inference,
on the functional level,
we might talk about model evidence.

But at a certain point,
you cross a certain threshold leading to model selection of you altering your appropriate
perceptive pose,
and this becoming a self-fulfilling prophecy through active inference.

You then minimize the prediction error between your imagination,
I am moving,
and the action through reflex arcs,
the way that basically motor control is thought to work in active inference,
and predictive processing.

So the psychedelic angle would be,
as someone is dosing with these compounds,
their imagination becomes more vivid,
and they might show stronger ramping at first,
and maybe greater awareness of this patterns of mental simulation,
leading up to the action.

It might get stronger still as you dose more and more.

Also greater coupling between interoceptive and proprioceptive modalities,

increasing the gain on this positive feedback between mental imagery and feelings of urge.
And so in a way,
this would be an enhancement of the willing aspect of agency as you move up this dose response curve.
But then you get to a certain level,
and you start to get into this psychedelic regime where your thought becomes highly creative,
and then your directed willing might start to go down.
Potentially, the idea is you might find other things competing for your awareness,
different creative possibilities.
So you're now imagining nature very vividly,
and this is competing with you rehearsing,
will I move or won't I move?
And you might get weak and less predicting I'm about to move.
And as you keep up with the dose,
these inferences,
you directing a certain magic about one move or one move
might become less and less extremely high doses.
You might get a compromise of these mechanisms of agency,
of this conjoining of your interoceptive and your proprioceptive predictions,
of your stream of consciousness being pulled along by value-driven policy selection
via sophisticated affective inference.
So the reverse direction is this,
in terms of this paradigm potentially informing psychedelics,
could be that one source of hallucinations or delusions
or atypical altered consciousness and cognition
would be that if you take the basic micro-mechanics of agency
and you compromise that.
So you're beginning to somewhat imagine an action,
but it's not well integrated into some causal confolding,
you could come to this anomalous inferences
about what's causing these basically,
the different endogenous activity in your brain,
the different events that could lead to motorization,
if these aren't being integrated into a self-consistent,
coherent causal unfolding,
one of the ways that you could try to make sense of this
would be a confabulation of something like a hallucination.
And so this could potentially also explain
some of psychedelic phenomenology

and potentially clinical neuropsychiatric phenomenology
with things like schizophrenia
as potentially being a function of disrupted agency mechanisms.
So that's currently the area of focus.
And there's going to be a preprint just came out on that called
On the Degrees of Freedom Worth Having.
Psychedelics as a Window for Understanding and Expanding Free Will.
I believe that's what I called it.
Could you maybe unpack a little bit?
Also, apologies if there's any strange stream hiccups,
but could you maybe unpack what the alien
or extraterrestrial was doing
and how does the confabulation relate to a narrative understanding of action?
So in this one,
so we're starting out just like imagining your hand moving
or imagining being in nature.
And then same here as the dose goes up
and it goes up
and you might start to get some more intense interoceptive sensations.
The 5-HT_{2A} receptors,
which heavily mediate the effects of classic psychedelics,
they're highly expressed in the interior insula,
the top of the interoceptive hierarchy.
And so you keep going up,
the sensations become more intense,
imagination becomes closer to reality.
And so here,
like what I'm calling the heroic dose
in this extreme dose regimes,
you're starting to get,
for instance,
like your ownership of the actions
are starting to get undermined.
So here I'm depicting,
like you find your hand moving,
but you're not accounting for
your willing of the move
being the thing.

You're not saying,
oh, I was rehearsing it
and it felt this way
and that was part of the causal stream of action.
You just found yourself moving.
And so one way you could make sense of this
or confabulate around this here
is a hand,
like it's a feeling of the hand
pushing down on your hand from without,
just an example of a potential confabulation
or hallucination someone might have.
And this one here,
it's expanded
and it's this basically,
someone,
it's not even your hand anymore pressing it,
but an alien's pressing it.
It's just an example of,
for instance,
so psychedelics,
the classic psychedelics
will act in these 5-HT_{2A} receptors.
They are heavily expressed
on these deep pyramidal neurons
of cortex's six layers.
The layer five will basically,
they're big neurons
that loop with the thalamus
and they let you form
these broad synchronous complexes,
which in predictive processing
are thought to encode your beliefs,
some subset of which
would be your experience.
And so if you increase the gain on these,
the idea is that
you might have enhanced perceptual vividness,

but because though,
you've done a radical alteration
to,
of multiple kinds,
to like the normal regime
of processing in your mind,
the normal ways you would model,
even though you're experiencing things intensely,
this would potentially be
of an atypical variety of experience.

And so here,
it's the intense experience,
the intense perception,
apperception of the alien,
it's a counterfactual alien,
it's not actually an alien,
but that would,
that would be an example
of an anomalous inference
you might make.

And it has some relation
to what's happening
and the fact that
maybe there was some
motor rehearsal going on,
but you're not able
to keep track of it.

And so,
and here,
this is,
you're actually having
phenomenal consciousness
in each of these moments,
but you don't have conscious access
unless in these iterative moments
of experience,
you can somehow get callbacks
where things can be contextualized

within some sort of causal unfolding.

The claim would be
that basically this would be
a way of separating out
phenomenal consciousness
from conscious access,
that there's things
that can happen
as you're generating
these qualia states,
as estimates of what's happening
of system world,
system world,
system world,
specifically your sensorium
as you think it is
based on your current engagement
or in counterfactual circumstances
and that this,
that's the stream of consciousness,
but that not all
of the frames of experience
are accessible
or rememberable.

And so,
in this case here,
you know,
maybe there was some rehearsal,
but you're not keeping track of it.

And then,
the best causal account
you were able to come up with
was that an alien made you do it
and you might have even forgotten
that this was you
or maybe you do remember it,
but it would need probably
some more contextualization

or not.

It's unclear.

yeah,

so that would be the basic ideas

that psychedelics

would be increasing

consciousness level

or the vividness

of your experience,

but potentially undermining

the coherence

in terms of causal unfoldings

or your usual priors

by which you structure the world

in the spirit of

Robin Carrad Harris

and Carl Fristen's Rebus model,

which I'm currently discussing

in the context of a broader framework

that I call ALBIS.

but,

yeah,

I don't know,

did that help at all

or,

like,

the alien?

Yeah,

it also,

it reminded me of what you said

about the different levels

and their interdependence

and mutual constraints.

so it's like a sound is heard

and potentially in one state

of consciousness

or mind or brain,

it's 98% likely to be one thing
and option B and C
are 1% likelihood.
And so it just goes forward
with as if option A,
like,
as if I'm in control.
but then some things might happen
where,
like,
that 1% chance
that it is an alien
gets accepted
and sort of passed forward
and then that might lead
to the experience
through the generative model
of something that,
like,
was a distributional,
so it's like a change
in quantitative distributions
that leads to a qualitative change
at increasingly higher
and categorical
and semantic experiential layers.
Beautifully said.
And it would depend
on your cultural priors.
So whether we,
you would call this,
like,
a relaxed belief
or a strengthened belief,
I'm not sure.
But,
you know,
for instance,

you might not be seeing,
like,
if you,
you know,
or an indigenous people
of South America,
for instance,
you know,
they might not see an alien.

It might be
related to something
having to do
with their spirituality
and their,
whatever available myths
you have
would then go into
basically this
interpretation.

Your best guess,
which is always wrong,
sometimes useful,
but your best guess
of what's going on
will be different
depending on,
you know,
what they would call
the set and the setting,
but also where
the setting
is the broader cultural,
the setting
are also broader
to the culture.

Not everyone will be,
not everyone will come
to the inference

of aliens probably.

And actually,
that's been shown
that hallucinations
are culturally specific.

Yeah,
with which kinds
of extraterrestrial
morphologies
are seen
in relationship
to which are prevalent
in the media
and the stories
of a given region.

I don't know
all the details,
but one can imagine.

Yeah,
like not everyone,
I mean,
like you could like
say like
maybe there is
something about
like gray aliens,
which is like
a little more archetypal.

I don't know
if I'd say this,
but like if you look
at like the eigen dimensions
of like morphological
variation,
what's relevant to us,
like you're just
like exaggerating
many of the dimensions

that would be relevant
for us engaging
in intersubjective modeling
or modeling of ourselves
even for the sake
of control.

So like hands,
like bigger hands
might be important.

You know,
where is like
attentional inference?

Like where is this
person attending to?

Where am I attending to?

So you don't want to know
the heads and the eyes.

That's going to be
really crucial.

Yeah,
that's a far out story,
but you know,
there could be aspects
of what,
that aren't just
through the medium,
but.

It's super interesting
and yeah,
not to go into
alien stream mode,
but thinking about
a large headed archetype
with very large,
dark eyes
and the long fingers
like you're kind of
hinting at.

it's like,
it's all pupil.
It's all cortex.
It doesn't have
the strength,
but it has a lot
of nimbleness
with its hands
and with its body.
And so like,
that's like one archetype
of,
of cognitive action.
I don't know.
And then what,
maybe there's a blob
and a shadow.
There's like these
different archetypes,
different little characters
in this comic book
of cognition.
Not that different
from a child's
affordances too,
right?
Like they're not,
they're not capable
of like moving things
around much,
but it's like their
intentional actions
are mostly like
hand-eye coordination
of like,
that's their purchase
on the world.
That's their grip.

But the,
you know,
for all this,
you know,
so this is the current
like focus
is how might psychedelics
inform our understanding
of the micromechanics
of agency
and how might
the micromechanics
of agency
inform our understanding
of what psychedelics
are doing.
That's currently
one of the things
I'm focusing on
and about to
do some experiments
on at the
Center for Psychedelic
and Consciousness Science
I'm working at
at Hopkins.
but the fundamentally
and upstream of this,
the idea is that
these studies
which purportedly
show that
there's no free will,
they show no such thing.
It's very ambiguous
what they're showing
and I believe
if you do

an active inferential
handling of them,
you'll actually see
that there are several ways
in which consciousness
is contributing
to causal streams
leading to action,
which would be
having
some of the varieties
of free will
worth having.

Is it arbitrary
degrees of free will?

No.

Are there some varieties
of free will worth having
that we don't have?

Can you just fly all of a sudden?

That'd be nice.

That'd be worthwhile.

We can't do that either.

But if you want

your beliefs

and desires

to be able

to meaningfully said,

to be able

to be said

to be

meaningful,

efficient core screenings,

maybe even maximally

efficient core screenings

of what is happening,

I think we do have that.

And I think

you can find this
even within
these Labette paradigms.
if you didn't,
I think you still
could have many
of the varieties
of free will worth having
because many people
have criticized
and said,
oh, just moving your hand
without a reason,
that's not really agency.
That's not what we want
from free will.
We want meaningful decisions.
Sure.
But I think
even for these studies,
which people often
will trot out
and say,
look,
my brain made me do it.
Even then,
I don't think
it follows
that
they show
that our consciousness
is epiphenomenal
through and through.
And I think
active inference
basically shows
some of the particular ways
that consciousness

is indeed causal
even in these circumstances.

And I think
that's a very big deal.

I don't know
how well it's surviving
the replication crisis,
but there's plenty
of studies
showing that
some undesirable
consequences
of people
not believing
in their own free will
or their own agency.

Whether or not
that works,
I think there is
a kind of
this face-valid account
of this
and that like
kind of like
Dumbo's magic feather.

So like,
you know,
this elephant,
I'm taking this
from Dan Dunnnett
from his book
Freedom Evolves,
but this elephant
has a feather
that the crows
told him
will help him fly
and he had his big ears

and he used him
to fly when he had
the feather.
But it was because
he believed.
Whether or not
believing in free will
gives you free will,
not believing
you're a meaningful cause,
that becomes true
just by believing it.
So,
you know,
I think it's
quite dangerous
to take
basically
something where we have
something precious,
something that basically
would empower us
to pursue our values
and to say
based on
the set of studies
and using the authority
of science
telling people
that they are not
meaningful causes.
I think that's
extremely pernicious
and I think
active inference
can,
there's already
great work being done

within the
neurophilosophy
of free will
pushing back against this
and I think
active inference
can help
showing people
what are the
meaningful degrees
of freedom
that they have
and how can they
be preserved
and expanded.

Wow,
very interesting.
Do you have
any more slides
to go through
or may I ask
some questions?

No more slides.

That was it.

Okay,

so

those who are
watching live,
please feel free
to ask a question.

Otherwise,

I want to

pick up

near the beginning

when you

spoke about

finding oneself

in a stream

of action.

So what

is happening

when we find

ourselves

in a stream

of action?

And I

want to hear

what you have

to say

and then

take this

to some

questions

about memory

and free will.

But when we

are suddenly

aware,

like,

whoa,

I'm on a run.

I mean,

I guess I did

make prior decisions

about deciding

and so on,

but like,

whoa,

now my leg

is moving forward.

So the,

I imagine

this varies

a good amount

like as a

in,

in
inter
and intra
individual
difference variable.

So like both
between
and within
people,
you're going
to get a good
amount of variance
on this,
I think.

But the way
Heidegger described
it in terms
of this difference
between like
describing like
hammering and
when you're a
skilled craftsperson,
it's the hammer
is in your hand
and it is
ready at hand
and your mind
could be elsewhere
or people might
have the phenomenology
of this like
on a run
or on a drive,
you might just
find yourself home.
You can listen
to an audiobook,

you'll barely
know,
you're just,
you have this
effortless mastery
of the task set
and it's
only when
something goes
wrong that
things are
brought into
awareness.

And so
the,
it's only,
and so then
you get this
motion between,
so here's the
hammer just
in your hand,
ready at hand
to present
to hand
and now
it's in the
foreground.

It was initially
in this unconscious
or semi-conscious
background.

And so
like if someone's
running,
like for instance,
they could largely
be in a similar

state of effortless
mastery,
especially because
you know,
just natural
pendulum motion
gets you like
a lot of the
control,
that combined
with some
spinal pattern
generators,
you know,
it's,
and little
other tweaks.

It's like
you barely need
to do anything
once you,
once you've been
trained up.

But let's
say you're
trail running
and,
or let's
say you run
and then
you step on
something.

The idea
would be
that this
violation
of your
expectations

would cause
basically
prediction error
and that
this prediction
error,
this thing
that was
different.
So,
okay,
so within
predictive
processing,
part of
one of
this marriage
between
Heideggerian
phenomenology
where things
just become
unconscious
and you only
become conscious
when you're,
something's
wrong,
you might,
while being
careful not to
conflate levels
of analysis,
you could get
a good amount
of this just
from predictive
processing

with a dual
tier architecture.

Like,
if basically
you're really
well trained
up,
you can explain
things,
you can basically
handle things
just with these
efficient,
unconscious
couplings,
unconscious
inference,
as in a bunch
of reflex arcs
being skillfully
deployed,
doing most of
the things you
need to do,
and whether you
want to call
this modeling
or not,
you know,
it would be a
matter of choice,
I would call
it modeling,
but you wouldn't
need to become
aware of it
unless something
was wrong,

and so then
the idea is
this lower down,
it's not handled,
you don't predict
it,
you don't explain
away the prediction
error,
it makes its way
deeper into the
brain,
into,
it becomes
more broadly,
the deeper
portions,
of course,
it's
more explicit
modeling
and conscious
modeling,
this
processing
will be
greater,
more capable
of accessing
these richly
connected sub
networks,
which could
support basically
this larger
scale integrative
simulation,
some aspect of

which would be
your experience.

I don't know
if that got it
with it all,
but.

Yeah,
awesome.

So we're,
we're driving
back towards
free will
because I
really liked
how you
approached it
in,
in this
conversation.

So I want
to think
about three
concepts,
memory,
free will,
and experience.

What are
their Venn
diagrams?

Like,
can we,
of things
that we
remember,
did we have
free will
for them
or not,

more or less
than expected
for things
that we
experienced?
Do we
remember them
more or not?
Could there
be something
that you
didn't experience
but you
remember
and had
free will
on and
so on?
Like,
how are
memory,
free will,
and experience
bound together?
That's a
fantastic question.
I don't,
so what's
coming to
mind,
that's a
whole other
conversation,
but what's
coming to
mind for
me at the
moment is

that there's
a partial
definitional
aspect of
a connection
between
free will
and experience
and that
what people,
so people
want different
things.

I wouldn't
call free
will one
thing.

In fact,
I actually
think there's
a certain
wisdom in
just like
the language
of it.

Like,
people want
both like
the willing,
they want
the agency,
and they want
the degrees
of freedom
in terms
of they
want to
be able

to pivot.

They want

to be able

to have

what they

do be a

function of

their past

and being

constrained by

their memories

and their

meanings from

the past,

but they also

want to be

able to

break free

whether in

the face

of something

new or

just encountering

something new

or to

explore.

And so

having that

freedom for

exploration is

also going

to be a

part of

agency,

both within

a context

and across

time.

But the
memory is
going to be
important
because without
the memory,
that's the
constraint based
on values,
based on you.
There's many
aspects of the
self,
for instance,
some of the
most essential
ones and maybe
some of the
gold standards
of free will,
some Frankfurt
mesh theory,
it's like this
interface of
you both want
things and you
want that you
want them,
where the
wanting that you
want is based
on this
integrative and
narrative type
self.
And so
that would
be a way

in which
memory would
very strongly,
where free
will would
be a
function,
both of
experience that
you want your
consciousness and
your experience to
be part of the
causal stream leading
to action,
part of your
valuing,
your affective
inference,
you want that
to be influencing
both across time
and in the
moment.
And so the
experience would
be essential.
It's not clear
that many,
like Dennett has
argued that free
will and consciousness
must always be
handled together.
without memory,
you also,
though,
it's not you

either.

I think there's
an interesting
thing,
though,
in terms of
like you're
alluding to
like basically
connections between
causation and
memory.

And I believe we
are better capable
of remembering
things when we
can take them
up into a
causal unfolding.

This gives,
this basically
affords a
representation
with a more
of a sparse
structure of
salience,
of this led
to this,
led to this,
and then you
hold that
causal graph
in your mind
as basically
the core of
your generative
model,

and it seems
to be like a
highly efficient
way of remembering
things is to
come up with a
causal account
of them.

And so if
something is a
product of your
agency,
in a way,
that is
automatically
structuring things
in this causal
fashion,
and so there
should be a
connection between
having agency
and free will
in a circumstance
and being better
able to remember
things.

I don't know
if that's the
case,
but it
seems like
some things
from education
theory suggest
with respect
to active
learning,

but I could
see that going
a lot of
different ways.
That's a deep
question,
and that's what
came to mind,
but what do
you think?
Well,
the last piece
you brought up
about the
sparsity,
kind of the
sharpness on
the sparsity
of the causal
story,
and then
things that
have a sharp
and a sparse
causal
architecture
as experienced
with our
very limited
working memory,
they're like
pre-digested
and therefore
very apt
to be able
to leave
certain kinds
of traces,

and then maybe
something could
leave a trace
that didn't
have a perceived
sparse causal
architecture in
the moment,
but that
might be
an example
of a memory
that's hard
to recall,
or that
might be
cleaned up
in post-production
and maybe
very susceptible
to being
influenced
because subsequent
updates could
try to draw
attention to
part of that
uncertainty,
and if there
was just a
loose memory
of many things
happening at a
scene,
asking even
a question
about part of
it could start

to sharpen
that aspect,
and so
it's an
interesting
question,
and also
kind of like
the questions
like,
what if you
had some
sort of
suffering,
but you
didn't
remember it,
or in
between
moments,
or when
you were
sleeping,
you were
having some
sort of
negative
experience,
so just,
I mean,
what is
happening with
the mind
and the
brain?
the brain
seems to
be

continuous,
but then
we have
these more
flashes and
discreteness
and shifting
of mental
attention.

Go ahead,
go ahead.

I just
like the way
you're
describing
this,

I feel
like it's
like a
really good
case in
point of
the answer
to do we
or don't
we have
free will
should always
be,
what do
you mean?

And the
answer is
always like,
the way you're
showing these
different processes
could either go

together synergistically
or come apart,
it's like,
how many
meaningful degrees
of freedom do
we have in
what circumstances
or in what
senses do we
have,
how much of
which of the
varieties of
free will
that we want
in different
circumstances?

So it's like,
the answer is
there or isn't
there free will.

It seems like
much more
interesting is
getting into
the details
and useful.

It's like getting
into the details
you're talking
about right now
with like the
different ways
that memory
and experience
and choice
can all go

together or
not both
within and
across time.
Like that's
and then that
answer is like
I guess you
could say
that's like
that's the
real problem
of free will
and then that

interesting.
Yeah,
like when the
experiment of
pushing the
button is
presented as
being about
the exertion
of decision
making in
the moment
rather than
the
memory
or the
recalling
of something
that might
have been
different in
the moment
those are

quite different
settings and
it could be a
total different
neurophysiology
so and also
what you said
about how
this sort of
tension between
having a
connection to
the past and
the constraints
that it provides
but also the
ability to
choose and I
guess that's
what's explored
in these
varieties papers
so I wanted
to not just
ask like what
are the
varieties but
just specifically
asking about the
similarities and
differences with
bodily action
what are the
varieties of
free will we
might have with
respect to
bodily action

like the so
called voluntary
and involuntary
bodily processes
and then mental
action where
inactive inference
we can model
regimes of
attention and
metacognition and
thought as like
action policies
on mental
states so what
kinds of free
will are out
there for the
body and what
does that say
about what might
be a cognitive
degree of freedom
or not like you
know just choosing
what we want
so in terms of
body actions
if we're dealing
with something
highly rehearsed
then it
seems like
that could be
an example
or something
where it's
like

you can model
that the
predictive
processing
like the
hammer strike
and the
loop is
closed
before it
ever even
enters
like the
brain
it's in
the
the
the
cns
and the
reflex
arc
but doesn't
have to
even make
it up
top
before your
body moves
that's just
after the
fact that
you're aware
of it
but then
if you're
doing
something

like
and so
then like
at this
other
extreme
of
abstraction
you might
have
something
like
planning
reaching a
distal goal
where there's
many steps
involved
and you need
to come to
some sort
of sequence
of the
steps
and along
the way
actually I
don't know if
that would be
the extreme
but that would
be
like there's
certain things
that have to
happen to some
owners like
at the extreme

of abstraction
it might be
something like
going to grad
school
something like
that
or you
know
too real
getting a new
job
too real
ouch
but like
that itself
there's a lot
of things
that have to
happen before
other things
a lot of
dependencies
and a lot
of multiple
realizability
like there's
always multiple
realizability
like you know
I go to reach
for this glass
I could have
reached for
like this
I could have
done a weird
thing and

gone like this
or like
you know
I could have
pinky out
pinky in
you know
I prefer
pinky out
but like
there's all sorts
of things
that could have
done there
it is
multiple
realizable
but the
multiple
realizability
of something
like you
know
going to
college
going to
grad
school
that's
even
more so
like the
number of
ways you
can go
into it
and then I
guess I'm

thinking like
in this
middle regime
of something
that's still
like very
clearly
meteoric
but would
be novel
in that
like fine
grained
motor control
like I'm
learning for
the first
time like
I'm doing
a tea
ceremony
or I'm
learning
calligraphy
or I'm
just learning
to knit
or something
like there
it's the
fine-grained
things are
now being
monitored as
present at
hand and
you actually
need to

use your
sophisticated
inference
machinery to
help scaffold
this basically
the acquisition
of skilled
motor
abilities of
skilled the
capacity for
skilled motor
performance
techniques
but I
would also
I think
suggest that
the whole
way through
it's always
built
experience
once it's
being co-opted
or controlled
and its
controllability
is always
modal
in a
motoric
fashion
so one
of the
claims I
make in

this paper
turns out
it's called
pre-motor
theory
but that
basically
like
through
partial
expression
of a
motor
command
or a
motor
prediction
the
efference
copy
from that
the
associated
sensations
can be
used
as a
means
of
top-down
attentional
selection
could be
used
as a
means
of
attending

to
memory
in
terms
of
working
memory
you can
use
these
partially
expressed
motor
predictions
not enough
to actually
cross
threshold
but use
them to
basically
as a
source
of
control
over
dynamics
of
where
your
awareness
is
going
and
what's
going
in
or

out
of
your
awareness
and
the
claim
I
make
is
actually
that
this
is
exhaustive
like
this
is
the
only
lever
you
have
is
basically
whatever
is
built
off
of
the
control
hierarchy
off
of
skeletal
muscle

it's
the
claim
but
the
idea
would
be
like
even
though
talking
about
something
as
abstract
as
you know
going to
college
or grad
school
there's
still
going to
be
some
sort
it has
to
cash
out
as
modal
you
you're
seeing
yourself

in
the
different
contexts
even
for
the
abstract
like
the
most
like
justice
you
still
to
experience
it
it's
still
going
to
have
some
sort
of
thing
here's
justice's
scales
here's
the
courtroom
here's
seeing
this
memory
from

your
past
where
justice
was
served
here's
you
feeling
this
uneasiness
or
this
righteousness
so
the
idea
I
would
claim
is
that
any
kind
of
control
on
any
level
will
always
cash
out
notoriously
but
there
are
meaningful

differences

in

terms

of

whether

you're

engaging

with

something

which

is

involving

these

immediate

fast

feedback

loops

it

might

be

multiple

realizable

but

less

so

or

something

which

the

affordance

structure

is

more

being

played

out

in

terms

of
your
capacity
to
control
and
relate
to
it
in
terms
of
these
even
more
multiply
realizable
high
level
plans
of
all
the
things
situations
you
might
be
in
that
would
be
related
to
achieving
high
level
goals

or
abstract
goals
I
don't
know
if
that
makes
sense
well
one
one
thought
on
that
is
it's
like
a
jigsaw
puzzle
so
there's
only
one
way
to
assemble
it
and
it's
kind
of
like
there's
therefore
this

bundle
of
trajectories
of
different
realizations
of
how
one
is
going
to
assemble
it
like
there's
sort
of
what
happens
on
the
table
all
the
combinatorics
of
the
way
to
assemble
it
and
then
there's
like
all
of

the
real
life
combinatorics
about
how
many
minutes
am
I
going
to
spend
playing
this
jigsaw
and
then
I'm
going
to
go
get
up
and
do
other
stuff
and
so
the
multiple
realization
of
the
jigsaw
where
it

does
all
kind
of
come
back
to
being
completed
is
one
thing
and
then
yeah
you
brought
up
how
when
things
are
even
the
outcomes
are
unknown
or
there's
paper
and
also
the
vivid
imagining
of
different
future

states
what
is
the
regime
of
attention
that
might
for
somebody
who
wants
to
be
an
athlete
help
them
make
practicing
and
being
on
that
path
more
natural
so
how
specific
and
what
modalities
that's
kind
of
what

it
made
me
think
of
but
I
kind
of
want
to
keep
on
going
on
this
mental
action
route
has
there
been
anything
like
a
mental
or
a
cognitive
libe
experiment
like
where
the
instructions
would
be
like

take
a
couple
of
breaths
and
just
you know
get up
and walk
around
or
whatever
just
take
a
couple
breaths
and
then
identify
the
moment
on
the
clock
when
you
chose
to
think
about
x
a
pure
mental
on
mental

assessment

so

before

I

forget

I

actually

think

what you

just

mentioned

the

optimal

attentional

engagement

for

something

like

a sport

or some

performance

actually

that's

a key

example

of

through

skilled

mental

action

or not

through

the

skillfulness

of

your

mental

actions

having
more
or
less
freedom
where
like
for
instance
if
you're
choking
for
instance
you
do
it
wrong
June
Tani
has a
good
paper
on
choking
and
active
inference
and
so
if
you
get
that
wrong
that
would
be

like
your
free
will
being
undermined
by
your
mental
action
not
being
well
calibrated
in terms
of
where
you're
positioning
your
attentional
selection
for
optimal
grip
or
optimal
engagement
with
what
you're
doing
in
terms
of
this
purely
more

purely
mental
action
there's
a couple
there's
different
versions
of it
that have
been
run
some
of
which
I
think
would
get
a
little
bit
more
to
that
and
kind
of
what
you
described
in
terms
of
Ari
Maus
has
one

paper
called
decisions
deciding
or decisions
that matter
and so
basically
it would
be
choosing
between
charities
and so
in this
charity
choice
you
actually
don't
see
this
same
type
of
ramping
and
I
think
this
support
to
some
degree
the
model
I'm
suggesting

and
that
like
if
this
ramping
is
a
buildup
of
activity
in your
action
oriented
body
maps
cross
coupling
with
your
interoceptive
body
maps
this
feeling
of
urge
and
mental
stimulation
it
makes
sense
that you
would
have
that
for

something
like
pressing
a
button
but
if
it's
something
like
which
charity
I'm
going
to
donate
to
you
might
not
get
that
same
tight
coupling
and
feedback
and
so
I
think
that's
some
evidence
in support
of
that
but

I
think
the
particular
thing
you
brought
up
there's
these
and this
is a
direction
I'm
looking
in
beginning
to look
into
there's
these
closely
related
waveforms
to the
readiness
potential
the
event
related
negativity
and
the
SPN
stimulus
preceding
negativity
and

I
believe
some
of
them
would
be
preparing
for
information
that's
coming
and
then
acting

I
think
that
would
be
the
ERN
or
the
SPN
the
stimulus
preceding
negativity
is
you're
preparing
to receive
information
that's
possibly
salient
but

there's
no
action
required
so
I
think
that
might
be
potentially
you
could
think
of
all
these
things
though
inactively
as
having
they
look
very
similar
electrophysiologically
and I
would suggest
that they
are
all
fundamentally
similar
and that
what
they're
using

whether
we're
talking
about
patterns
of
attentional
selection
or
specifically
orchestrating
mental
simulation
to
act
or not
in terms
of an
overt
action
that in
all these
cases
it is
grounded
in the
modalities
and this
is
basically
understood
implementationally
as recurrent
activity
in different
places of
a control
hierarchy

over body
maps
or
in terms
of
algorithmically
and functionally
an accumulation
of model
evidence
for shifting
your
proprioceptive
pose
now before
an overt
action
but for
talking
mental
action
similar
story
but if
partially
but if

you're
if the
model
evidence
I guess
would be
a slightly
different
model
but
it's

if
what
you're
imagining
is
your own
you're
imagining
your body
being in a
different
state
that would
allow you
to orient
differently
to bias
your sensors
differently
to move
your attention
around
differently
to do
different
precision
weighting
but
you
you
don't
imagine
it
vividly
enough
that it
will
actually

drive
your
spinal
reflex
arcs
and that
but that
this
would be
what you
would
use
even
for
potential
purely
mental
acts
it would
still
be
co-opting
a control
hierarchy
off of
skeletal
muscle
and the
reason I
focus on
skeletal
muscle
is
because
you know
it's
designed
to have

basically
large
gross
actions
with
low
latency
and
it's
controllable
and
and
and
the
conditions
for
fine
grade
control
are
realized
because
you
can
actually
direct
your
sensors
using
your
effectors
with
a
large
scale
meaningful
outcome
that

can be
causally
modeled
and tracked
and controlled
and so
then the
idea
is that
basically
Vygotsky
first expressed
an idea
like this
I think
with respect
to
thought
as
inner
speech
where
first you
learn how
to speak
and then
through
partially
expressing
your
speech
you
learn
how
to
co-opt
this
process

for
a
stream
of
thought
as
inner
speech
whether
you
want
to say
that
exhausts
thinking
is
another
thing
but
inner
speech
would
be
part
of
a
significant
part
of
thinking
but
so
the idea
is
this
is
all
intentional

mental
operations
whether
we're
talking
about
shifting
around
basically
the
fine
green
adjustment
of an
overt
skilled
motor
act
or
guiding

imagery
and
imaginings
that
in
each
case
you have
the
initial
affordance
structure
the initial
control
levers
established
via

these
overtly
expressed
behaviors
and couplings
of the world
which then
later
by partially
expressing
them
gives you
something
you can
repurpose
for things
like imaginative
planning
or covert
attention
or I
would argue
any
intentionally
influenced
mental
process
maybe
ultimately
have
its
origins
in this
co-opting
of
motor
control
bold

claim
okay
so what
systems
yes
go ahead
I just
said
could be
wrong
not
financial
advice
not
not
extraterrestrial
advice
hardly
even
active
advice
it's
very
interesting
what
systems
rise
to this
capacity
and does
their
embodiment
matter
so like
there are
insects
with
tremendously

reduced
central
nervous
systems
one
could
imagine
a
computational
system
that can
engage
in
counter
factuals
but
it's
only
affordance
or
embodiment
is
like
one
you know
morse code
receiver
or something
like that
so what
systems
have this
capacity
and why

so I
definitely

don't know
the boundaries
but if I
were to
shoot from
the hip
on this
the idea
would be
the handling
would still
be of this
computational
neurophenomenology
or
Maury
neurophenomenology
but instead
of drawing
brains
we would
draw
a different
nervous system
or whatever
the
signaling
information
processing
system
is
we would
still give
it this
kind of
handling
where there
would be a

mechanism
there would
be a
function
there would
be some
intermediate
algorithmic
mapping across
these levels
and then you
connect this
to phenomenology
and so if
in this
phenomenology
you're having
basically goal
states being
represented
and the
representation
of these
goal states
is part
at that
level
you're
also able
to get
an account
of the
guidance
of action
selection
to bring
them about
then you

could say
that you
have
degrees
of freedom
in that
way
so if
we're
talking about
something
like
an
insect
you actually
sent me
a really
interesting
paper
recently
about
dual
systems
thinking
broadly
construed
including
for insects
where
and
this
it seems
like you
would need
at least
a
two-tier
hierarchy

where this
upper level
can
break free
of immediate
engagements
with the
sensorium
and where
this lower
level
would allow
for your
couplings
with the
world
and the
orchestration
of overt
action
selection
in terms
of actually
like
sensory
motor
couplings
of the
world
and then
a process
which can
basically
on top
of that
hierarchically
higher
that could

infer
these
state
transitions
but without
them being
expressed
and if you
have that
then you
could do
the types
of
counterfactual
processing
that you
would need
for planning
and constructing
causal
models
it seems
possible that
insects might
be able
to do
this
in terms
of
for instance
like in
that paper
you sent
recently
like
these
like
do

this
so
the
central
complex
has like
a lot
of these
properties
of the
hippocampal
system
it's
kind of
like it
has properties
of the
hippocampal
system
and the
thalamus
at the
same
time
but
and it's
giving you
like the
salience
map
used
for both
orienting
and moving
but
you also
get these
mushroom

bodies
which are
like the
stratum
which would
be basically
helping with
moment-to-moment
action selection
biasing it
in a value-driven
direction
and so
in this paper
you're getting
like this
dual
systems for
control
one which
seems to be
more immediate
and reactive
and the other
seems to be
governed by
slower
processes
and so
it's possible
that something
like this
so it's
possible that
this could
give you
the ability
to do

sweeps
of possibilities
of counterfactual
inference
where you
basically take
the beginning
of expressing
the motor
command
so the
insects
thinking
I'm going to
locomote in
that direction
I don't know
the evidence
well enough to
know if this
is the case
but it
could be
and then
so for
like a
rodent
you'll see
these sweeps
in the hippocampus
of like
there's a maze
and you'll
see like
predictive
of whether
mouse goes
left or right

you'll see
these place
fields representing
different points
in space
and in anticipation
of this
you'll see
these sweeps
ahead of activity
and then
where the sweep
is most robust
that'll be the
branch
that like
the animal
will go down
it'll go down
that's the
places
in theory
you could do
a similar
thing by other
means
maybe with
some homology
even in the
insect nervous
system
maybe you can
even do it
in an ant
colony
if you're
getting like
an inner

attractor
capable of
aggregating
across all
these different
finer grain
couplings
and engaging
in control
but so
basically
whatever
has
I can't
say
I'm not
going to say
sufficient
but the
claim I'll
make for
necessity
would be
some sort
of dual
tier
architecture
where the
lower level
gives you
the couplings
with the
world
and the
upper
level
gives you
counterfactual

affordance
and
integration
across
these
lower level
potentially
segregated
coupling
processes
something like
that could
do it for
you
and so
could insects
do it
potentially
I don't
know
could the
insect
colony
do it
maybe
the thing
where I
would place
a potential
constraint
is
it seems
like we
might need
an
architecture
to have
properties

where it
could
signal
fast
enough
where
these
inferences
quickly
enough
that it
can
stay
on top
of
its
evolving
action
perception
cycles
so in
terms of
its
surfing
uncertainty
it's
moving
through
the world
as
couplings
the
architecture
whether
we're
talking
about
a

brain
or
some
set
of
signaling
across
multiple
individuals
like a
hive mind
we want
it I think
to be
fast
enough
to
have
the
estimates
correspond
to this
sparse
causal
graph
what Benjo
would call
I think
independently
controllable
factors
of the
world
and
a
causal
world
model

where you
are an
agent
within
it
but it
seems
like
you
would
need
to
have
the
ability
to
model
independently
controllable
factors
and so
you
need
a
signaling
system
that's
fast
enough
to pull
this
off
and so
some
sort
of
small
world

network
architecture
would
be
likely
required
where
you
have
a
combination
of
local
modularity
but
then
synergy
through
having
some
central
hub
and
so
if
this
could
be
pulled
off
by
the
hive
mind
and
if
this
can

orchestrate
the
overall
swarm
to
do
things
quickly
enough
to
basically
have
purchase
have
the
right
grips
on
the
world
that
it
can
both
inform
and be
informed
by its
action
perception
cycles
then you
could
have
an
agency
architecture
which

by some
dimensions
we might
describe
in terms
of
free
will
that
dodged
the
question
before

I
respond
what
about
just
the
computational
one
what
if
it
was
a
computational
one
and it
was
instantiating
a
digital
twin
a
limited
or

comprehensive

digital

twin

of

itself

I

mean

I

personally

think

the

like

an

integrated

information

theorist

would say

absolutely

not

some

people

within

like

active

inference

communities

would say

no

it

has

to

be

physically

realized

but

I

think

something

like
a
digital
twin
or
some
sort
of
emulation
the
important
thing
for
me
is
that
there
is
a
set
of
dependencies
of
effective
connectivity
that
could
be
involving
virtual
processes
and
that
if
you
trace
out
the

patterns
of
effective
connectivity
of
kinds
not
necessarily
physical
not
necessarily
made
of
matter
that
if
that
is
entailing
the
modeling
of
a
system
and
its
relationships
with
the
world
and
where
within
that
system
world
model
it

has
this
kind
of
agentic
affordance
for me
that counts
but a lot
of people
would disagree
on that
but for
me I
think
you're
trying to
find the
maximally
explanatory
core screening
and sometimes
that might be
found at the
level of the
virtual machine
and not in
terms of the
physical
realization
and then
that's
actually the
most meaningful
cause
and then
if you
trace it

out there
that's
where you
would look
for it
and so
that could
be
in a
digital
twin
or
emulation
but a lot
of people
would disagree
with that
so when you
were talking
about applying
this to the
colony level
I thought about
like the
foraging
behavior
that's like
reaching out
of the
ground
and grabbing
a seed
and so
that's like
the enacted
foraging
trip
and then

is there
a counterfactual
I could
be foraging
or we
could be
foraging
at some
different
scale
underground
on the
other side
of the
markov
blanket
of the
nest
entrance
and then
what if
somebody
were going
to be
tracking
the
elevation
of a
forager
and they'd
be like
well they
go up
and down
in the
nest
but on
the times

where they
make that
decision
to forage
they always
go up
200
milliseconds
before
foraging
it's like
well yeah
that's where
the nest
entrance
is
so
without
going into
whether they
have will
or not
it's like
it's some
kind of
cognitive
motor
integration
when the
person
hits the
button
so then
the way
that you
frame this
as being
like not

a trivial
artifact
in fact
an interesting
outcome
of
aggregation
of
ERPs
across
trials
it's
like
how
couldn't
there
be
a
coherent
preamble
now I
guess it
doesn't
if we're
30 seconds
long
that would
be a
different
setting
but then
when we
connect it
to the
specific
mechanisms
by which
there

that do
have to
come
together
in order
for there
to be
a coherent
motor
action
it's
quite an
interesting
view
and I
wonder
how
those
who
have
interpreted
it
in a
very
different
way
feel
have you
spoken
or asked
to anyone
who
has a
very
clear
and
different
interpretation

of what
this
potential
means
I've
been
discussing
it
some
with
I forget
I've been
discussing
some
with
or a
good
amount
with
Uri
Moes
and
Aaron
Scherger
Aaron
Scherger
has
he has
multiple
views
but some
of them
do
disagree
and so
one of
them is
you in

theory
because
you're
averaging
across
multiple
trials
there is
an
interpretation
of
readiness
potential
as
artifact
and
it's
possible
that
sometimes
you
are
dealing
with
an
artifact
of
measurement
and
this
is
like
an
open
question
but
he's
also

for
instance
he's
testing
out
the
mental
simulation
view
also
another
interesting
hypothesis
he has
is
actually
some
of
the
evidence
accumulation
is
actually
fast
feedback
loops
with
muscle
spindles
and
actually
you're
getting
basically
the
urge
actually
a lot

of
it
is
actually
in
the
body
itself
but
one
interesting
thing
before
I forget
is
that
these
buildup
and then
threshold
crossing
I
believe
this
you'll
see
across
all
nervous
systems
where
you
get
action
selection
and
I
would

wonder
whether
this
similarly
to me
saying
this
isn't
just
a
brain
story
might
we
see
basically
something
like this
if we're
looking at
them
I think
in your
active
paper
you talk
about
stigmatic
coherence
but
through
some
sort
of
some
sort
of
measure

of
that
kind
you
might
see
generalized
action
potentials
for
large
scale
changes
in
switching
modes
of the
hive
mind
of the
colony
while I'm
speculating
I actually
also
in terms
of
multistage
hierarchies
I actually
wonder
whether
the
brains
like for
instance
the resting
state

networks
of the
brain
maybe
not
like
there's
different
granularities
and
parcellations
it can
be as
much
as
17
or
more
if
you
wanted
but
there's
some
solutions
that
are
like
maybe
you
know
so
seven
networks
are
most
well-known
but

there's
also
like
five
network
solutions
and
I'm
wondering
if
they're
actually
universality
classes
for
active
inferential
agents
that
you'll
actually
get
something
like
homolog
of
different
resting
state
networks
in
their
patterns
of
effective
connectivity
so
generalized

resting
state
networks
generalized
generalized
very
speculative
but
I think
we might
a few
interesting
things that
makes me
think of
and are
we making
cognitive
science
a creation
in our
own
image
if we
talk about
the generalized
hippocampus
and so
to what
extent
are we
or can
we
must
we
generalize
from our
own

action
perception
loop
is
putting
another
umbrella
which
the human
mind
is just
a
realization
of
and say
well
the
cognitive
domain
is bigger
than the
human
brain
because
it
includes
the
brain
body
and the
extended
niche
but also
includes
non-human
systems
like if
we want

a cognitive
science
that includes
a thermometer
or something
like that
like a
thermostat
is that
a
is that
a cope
because
actually
that is
still
the water
that that
model swims
in is still
human
and cognitive
like
I don't
know
but these
are interesting
pieces
I mean
it seems
like
it seems
like surely
what you're
describing
has to be
the case
and it's

like
and that's
like both
a good
thing
and a bad
thing
where it's
like we
need to
be able
to draw
on an
analogy
with something
we understand
like we're
always like
we're usually
making you know
sense of things
in terms of
other things
and so like
our own
experience
as this
base domain
in order to
give us
some purchase
like can we
do it
without it
you could say
like first
principles
thinking

but like
which first
principles
but then
there's like
all of the
like the
baggage that
could come
like the
disanalogies
like if
it seems
like we
can't avoid
doing that
maybe we
could
but I don't
know if
we want
to
but then
like we
really need
to be
careful
to track
the
disanalogies
like if
we're saying
generalized
hippocampus
like okay
wait wait
wait
what are the

disanalogies

here

what are the

unique

properties

that are

present

in the

target

domain

and not

the

base

domain

of like

the

analogical

mapping

you're doing

or vice

versa

yes

yeah

that's

fraught

what you're

describing

yeah

um

well I

think a

kind of

closing

area

just one

point that

I want

to pick

up on
I thought
it was
really
powerful
what you
said about
really the
social
consequences
and the
personal
consequences
of
interpretation
of
neurosciences
and in
some ways
even science
communication
around
empirical
results
like
the
idea
that
we
could
opt
out
to
free
will
and
that
in

a
world
where
there's
limited
degrees
of
freedom
you might
be
coerced
into thinking
you had
free will
or not
but if
there's
any
glimmer
of free
will
it can
be used
to
deny
and deceive
itself
so that
puts
research
in this
area
in
a
different
reflexive
position
than

research
on other
systems
so I
guess
just
how do
we
move
forward
here
while
drawing
on the
specifics
and being
respectful
of basically
the past
and the
present
and the
future
I mean
for me
I definitely
have a lot
of
soul
in the
game
and that
this is
actually
maybe the
biggest
thing
that

a couple
things
this might
have been
one of the
biggest
things that
got me
into
cognitive
science
was
a very
long
extended
freak out
about
free will
when I was
younger
where it
was just
I couldn't
see how
anything
could be
meaningful
without it
I don't
know if I
still believe
that
exactly
but
for me
it was
it was
very

difficult
to
not
believe
that there
were
like
this
sort
of
all or
nothing
thinking
about
free will
and then
going with the
nothing
Dennett actually
rescued me from
that with
Freedom of Olives
his book
but there
was a long
time where
it seriously
bothered me
and it had
real negative
consequences
on my
life
and I'm
not alone
in this
I've talked
to others

and they
also feel
that way
and part
of the way
this is coming
up in the
present work
with psychedelics
actually
is
so there's
this study
I'm preparing
for with
creative
professionals
who are
suffering
from burnout
and one
thing I'm
wondering
in terms
of like
people can
get burnt
out for
multiple
reasons
but to
the extent
that this
is continuous
with
burnout
is continuous
with

depression
or like
feeling like
you don't
have agency
in context
where you
want to
have agency
or feeling
like you
don't have
adequate
control
or something
where it
really matters
that you
do
that seems
like a
good
recipe
for
burnout
and it
seems
also
like a
decent
model
for some
forms
of
depression
where
if you
view

yourself
as not
being able
to be
a cause
in the
world
if you're
not able
to
meaningfully
work
towards
your
values
that
could
be
very
distressing
I mean
it could
also
be a
state
of
like
a
quasi
buddhistic
blissful
surrender
but
you know
there's a
question
are we
flirting

with
nihilism
there

is
a
question
and
the
other
question
an
eliminativist
on the
self
and on
volition
and then
just to
say
it's
okay
you're
not
real
your
volition
isn't
real
anyways
don't
worry
about
it
was
that
even
a

legitimate
intellectual
move
to
begin
with
I
would
argue
it
wasn't
so
yeah
I
think
this
is
the
stakes
here
for
people
sense
of
themselves
I
believe
they
really
matter
and
I
believe
they
have
both
social
and

clinical
and
social
consequences
existential
consequences
for
people
and
I'm
glad
that
there's
this
growing
community
of
people
in
the
neuro
philosophy
of
free
will
who
have
been
doing
a
really
great
job
pushing
against
some
of
these

excessively
inflationary
views
and
I
believe
that
active
inference
might
be
a
really
valuable
contribution
to
these
efforts
in
terms
of
providing
formal
models
and
models
that
can
go
across
this
marian
neuro
phenomenological
stack
or
computational
neuro

phenomenological

setup

and

actually

address

these

phenomena

in

enough

detail

where

not only

can

we

feel

meaningfully

satisfied

where

we

understand

our

own

agency

but

we

could

intervene

and

hopefully

for the

better.

Yeah

just

some

last

thoughts

on that

that's

very
powerful
and
the
mutual
constraints
and the
interdependencies
of the
different
levels
like
it
may
come
down
to
someone's
genetic
sequence
of a
receptor
or
some
kind
of
binding
affinity
with a
molecule
but
then
it
could
also
be
integrated
in

kind
of
a
full
stack
way
with
talk
or
action
or
changes
to
the
niche
so
it
gives
like
a
full
stack
yes
and
way
that's
not
I
think
a
dead
end
it's
more
like
a
starting
state

like
if
the
computer
weren't
working
and
then
there
was
like
the
hardware
and
there's
the
software
but
then
moving
beyond
just
that
type
of
analogy
which
is
very
powerful
about
burnout
and
about
the
different
cognitive
ways

that
we
might
get
there
and
get
into
it
and
get
out
of
it
and
so
it's
like
stoicism
or
whatever
plus
preferences
because
it's
not
just
the
neutral
stance
it's
not
just
a
descriptive
stance
on
burnout

there's
like
a
preference
there
so
Adam
if you
have
any
other
thoughts
please
feel
free
I guess
a
quick
thought
is
if we're
looking
and this
is like
to follow
up on
the
comment
where I
saying
like
these
somewhat
buddhistic
eliminativist
framings
of
agency

and I
said
like
are they
even
legitimate
I'm
going to
argue
no
they're
not
and
specifically
if we're
thinking
of
something
like
modeling
the
self
through
time
as
a
high
level
controller
if
you
engage
in
some
sort
of
practice
whether

it's
discipline
meditation
for a
long
period
of
times
or
you've
taken
like
a
very
high
dose
of
psychedelics
and
you've
basically
collapsed
the depth
of your
the temporal
depth
and counterfactual
richness
of your
generative
modeling
you will
not
find
yourself
there
you
will

be
gone
there
will
be
no
like
you
to
talk
about
and
you
might
call
something
like
ego
depth
but
then
to
say
that
there
never
was
a
you
process
to
begin
with
I
believe
that
is
actually

illegitimate

you've

basically

used

a

search

strategy

that

a priori

should

not

have been

able

to

find

the

self

and

then

using

that

search

strategy

you

then

concluded

it

didn't

exist

and

I

think

that

just

doesn't

work

so

in

this
case
to
quote
Dennett
again
individuals
their
narratives
their
persons
you
know
they're
real
patterns
and
but
they're
also
there's
a
reflexiveness
to it
where
their
own
belief
and
their
own
reality
is
an
important
part
of
their

causal
significance
and
so
you
know
I
think
it'll
be
good
to
have
some
nice
debates
with
like
different
members
of
the
active
inference
community
who
are
more
deflationary
on
things
like
self
processes
and
agency
because
ultimately

I think
this is
a thing
where
it's
going to
be
respectful
and
disagreements

I think
it really
matters
we get
it right
to the
extent
we can
thank you
Adam
for the
excellent
presentation
apologies
for any
internet
glitches
that might
have
happened
we are
a strange
loop
and this
has been
a strange
live stream
always

good
talking
to
you
Daniel
to
you
to
you
to
you
to

to

to
!
!

can
see
you
!

can
see
you
!
you

see

see

!